



Figure 2: Fast style transfer

In the style conversion process, the following steps are involved:

- Defining content and style representations
- Building the model
- Calculating the style
- Extracting the style and content
- Running the gradient descent
- Total variation loss
- Re-running the optimisation

The final stylised output with the source image rendered using the reference style image is shown in Figure 4.

The complete code demo is available at https://colab.research.google.com/github/tensorflow/docs/blob/master/site/en/tutorials/generative/style_transfer.ipynb.

Magenta

Magenta is an open source project aimed at exploring the role of machine learning in the creative process (<https://magenta.tensorflow.org/>). There are two different distributions of Magenta, as listed below.

- **Magenta – Python:** This is a Python library that is built on TensorFlow. It has powerful methods to manipulate music and image data. Generating music by using ML models is the primary objective of Magenta – Python. It can be installed with pip.
- **Magenta – JavaScript:** This distribution is an open source API. It enables the use of pre-trained Magenta models inside the Web browser. TensorFlow.js is at the core of this API.

Magenta-Python

The first step is to install Magenta along with the dependencies, as follows:

```
print('Installing dependencies...')
!apt-get update -qq && apt-get install -qq libfluidsynth1 fluid-
soundfont-gm build-essential libasound2-dev libjack-dev
!pip install -qU pyfluidsynth pretty_midi
!pip install -qU magenta
```

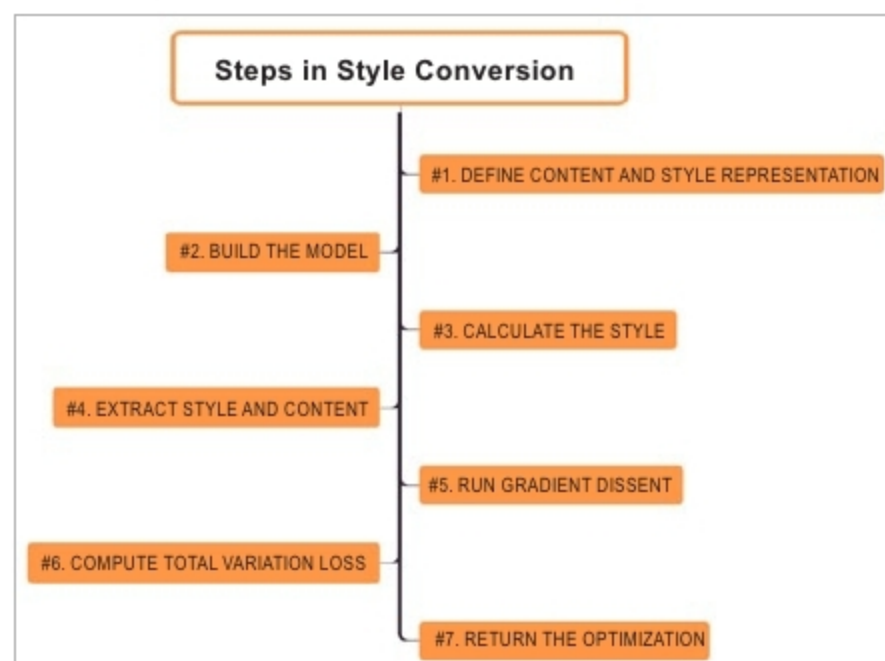


Figure 3: Steps involved in the styling process



Figure 4: Stylised output

Magenta fundamentally works on NoteSequences, which are abstract representations of a sequence of notes. For example, the following code snippet represents ‘Twinkle Twinkle Little Star’:

```
from magenta.music.protobuf import music_pb2
twinkle_twinkle = music_pb2.NoteSequence()
# Add the notes to the sequence.
twinkle_twinkle.notes.add(pitch=60, start_time=0.0, end_
time=0.5, velocity=80)
twinkle_twinkle.notes.add(pitch=60, start_time=0.5, end_
time=1.0, velocity=80)
twinkle_twinkle.notes.add(pitch=67, start_time=1.0, end_
time=1.5, velocity=80)
twinkle_twinkle.notes.add(pitch=67, start_time=1.5, end_
time=2.0, velocity=80)
twinkle_twinkle.notes.add(pitch=69, start_time=2.0, end_
time=2.5, velocity=80)
twinkle_twinkle.notes.add(pitch=69, start_time=2.5, end_
time=3.0, velocity=80)
twinkle_twinkle.notes.add(pitch=67, start_time=3.0, end_
time=4.0, velocity=80)
```